

How is a Blockchain Helpful in a Pandemic?

As the name suggests, blockchain is a chain of blocks that carry data (usually about any transaction) and hashes (created for the linked blocks). It is a unique method of storing data in a distributed manner, which also makes it a very secure system. This is precisely why a blockchain can help in many ways during a pandemic.

It was Stuart Haber and W. Scott Stornetta who first described a way of storing data using a chain of blocks linked together with cryptography. However, blockchain technology was improved by Satoshi Nakamoto in 2008 and implemented as a cryptocurrency called Bitcoin. Apart from Bitcoin, many financial services firms, logistics companies and social media giants are now trying to implement blockchain for their products and services.

Blockchain architecture

A blockchain consists of blocks (a

virtual data container) that contain data and a hash. The hash is created based on the data stored in the block, and this is done with the help of a hash creation algorithm. The block also contains the hash of its previous block. A block is created whenever any new data is stored in the blockchain network or any transaction is done (in case of cryptocurrency).

Blockchain uses a peer-to-peer (P2P) network and whenever a new block is created, it is linked with the previous one in the network by storing the hash of the previous block in it. This makes a blockchain secure, because if

the data of any block is tampered with, then it will change the hash of that block and it will no longer be connected with the blockchain.

Blockchain also uses a timestamp for every new data stored in a block and that prevents it from being tampered with or copied. While timestamping, the exact time of the stored data is stamped (stored) inside the block along with the data. If any change is made to the data, then the time also changes. This feature of blockchain makes it secure for cryptocurrencies, because if any transaction data is tampered with, then the time of transaction is also changed.

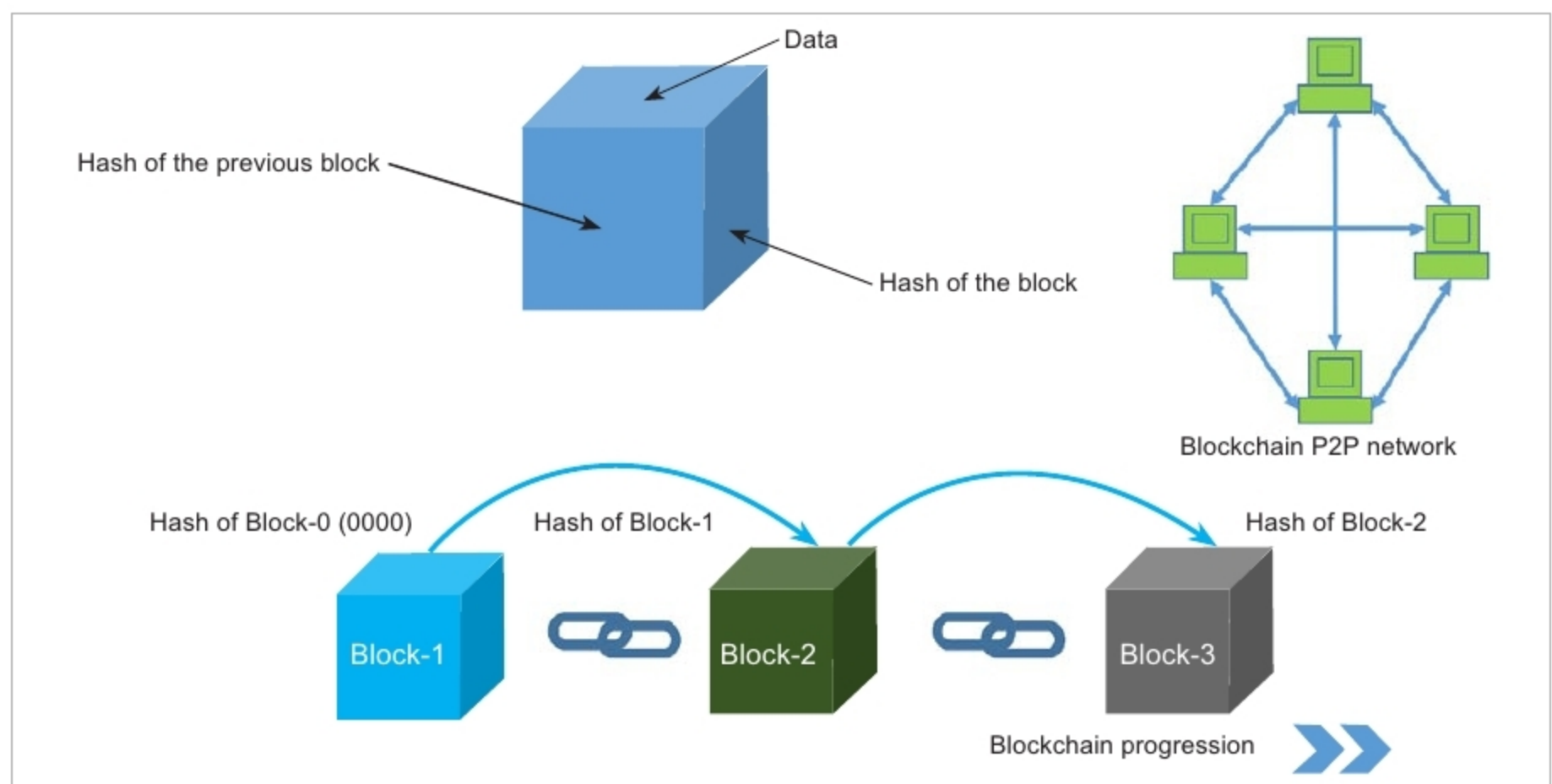


Figure 1: Blockchain architecture

